

241:

aabbccdd

$X \leftarrow 1$

$XY \leftarrow X$

$PPY \leftarrow XY$

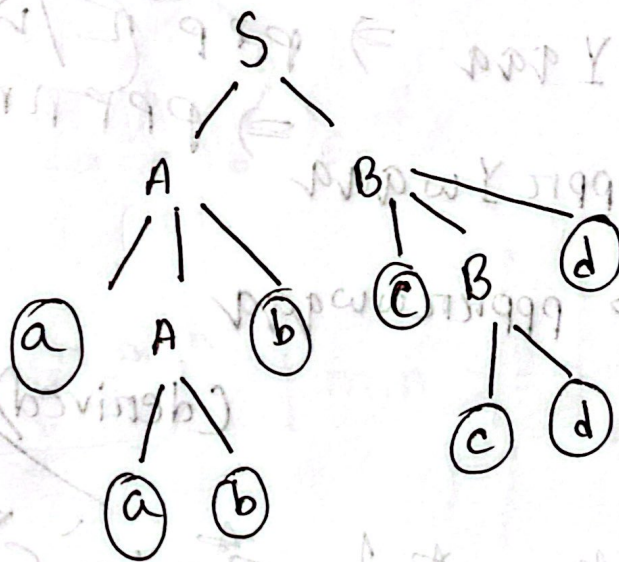
$PPPY \leftarrow PPY$

$PPPPY \leftarrow PPPY$

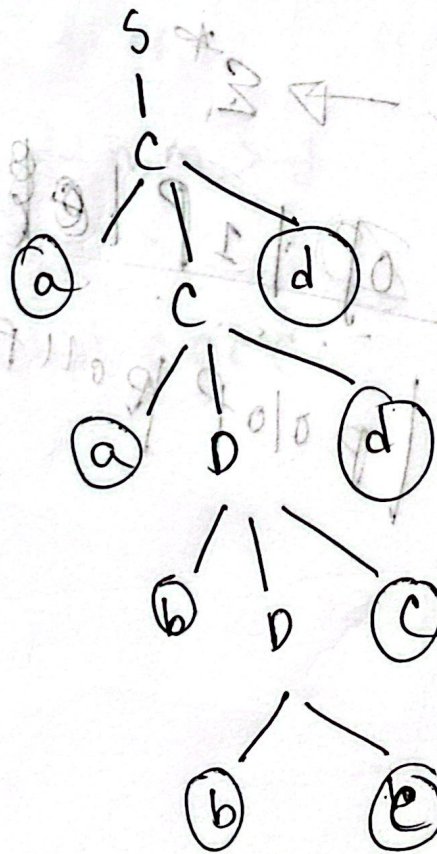
$PPPPPY \leftarrow PPPPY$

11

(a)



(b)



b. ✓

pppππwwqqa

Leftmost derivation:

$$E \xRightarrow{lm} X$$

$$\Rightarrow p \underline{Y} a$$

$$\Rightarrow pp \underline{Y} aa$$

$$\Rightarrow ppp \underline{Y} aaa$$

$$\Rightarrow ppp \pi \underline{Y} waaa$$

$$\Rightarrow ppp \pi \pi \underline{Y} wwaaa$$

(derived)

241:

2]

$$\{0 < n, m \mid \begin{matrix} n \\ m \end{matrix} \in \mathbb{N} \mid m > 0\}$$

(a) $P \rightarrow OP \mid 1P \mid \epsilon$

$S \rightarrow P110 \mid P010 \mid P011$

(a) $110 \rightarrow (a) \text{ begin}$

(d) $110 \rightarrow (d) \text{ begin}$

(d) $110 \rightarrow (c) \text{ begin}$

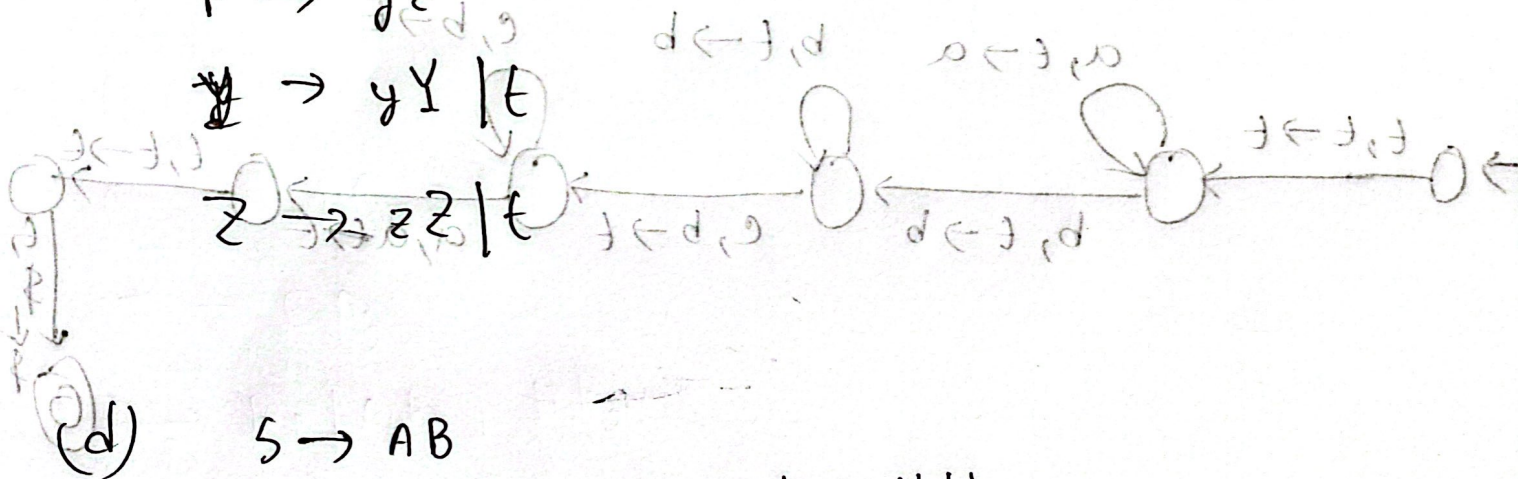
(a) $110 \rightarrow (a) \text{ begin}$

(b)

$S \rightarrow xzSp \mid \epsilon$

$P \rightarrow yz$

$\cancel{P} \rightarrow yY \mid \epsilon$



(d)

$S \rightarrow AB$

$A \rightarrow aAb \mid aaXbb \mid aaYbb$

$X \rightarrow aX \mid a$

$Y \rightarrow bY \mid b$

$B \rightarrow cB \mid cccc$

4]

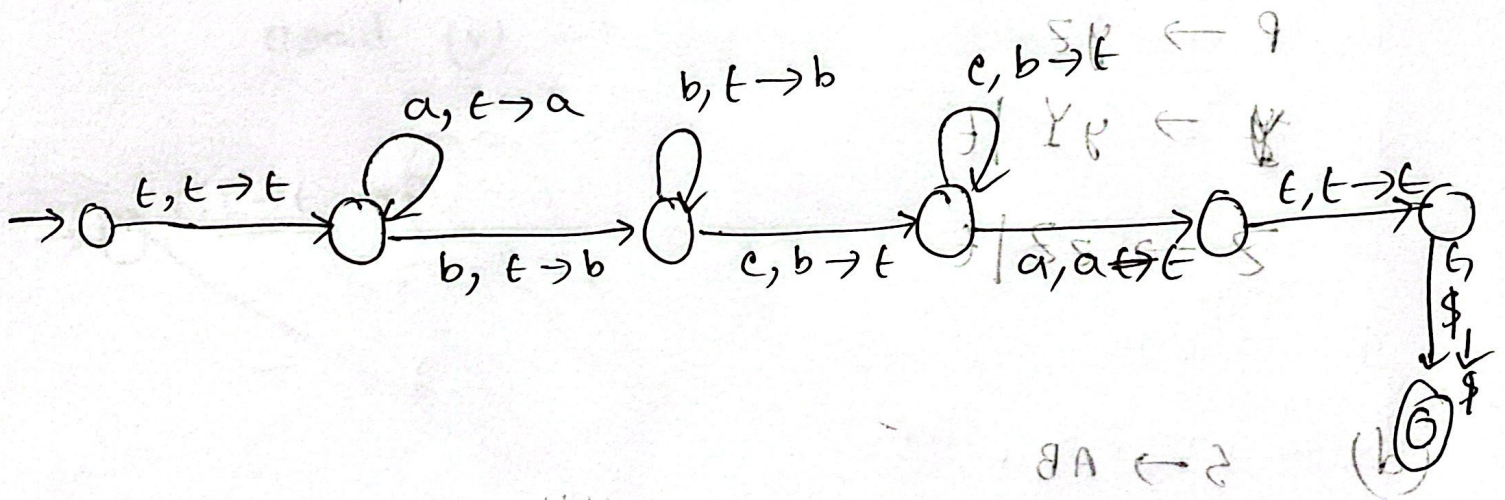
$$(a) L = \{ a^m b^n c^n a^m \mid m, n > 0 \}$$

read (a) $\rightarrow$ push (a)

read (b) $\rightarrow$ push (b)

read (c) $\rightarrow$ pop (b)

read (a) $\rightarrow$ pop (a)



$A \rightarrow aAp \mid aXp \mid aAp$

$X \rightarrow aX \mid \epsilon$

$Y \rightarrow pY \mid p$

$B \rightarrow cB \mid ccc$

⇒ switch (5): ~~case~~ 1: ~~STMT~~ CASE

⇒ switch (5): case 1: a+b case ~~EXPR~~: STMT CASE

⇒ switch (5): case 1: a+b case Int: STMT CASE

⇒ switch (5): case 1: a+b case 2: STMT CASE

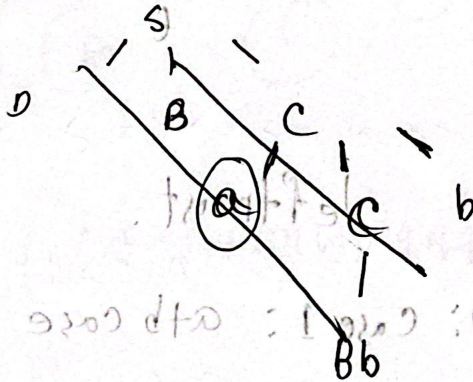
⇒ switch (5): case 1: a+b case 2: a++ ~~DF~~

⇒ switch (5): case 1: a+b case 2: a++ df: STMT

⇒ switch (5): case 1: a+b case 2: a++ df: a*b

⇒ switch (5): case 1: a+b case 2: a++ df: a*b

~~a 0 1 a a 0 b b b~~



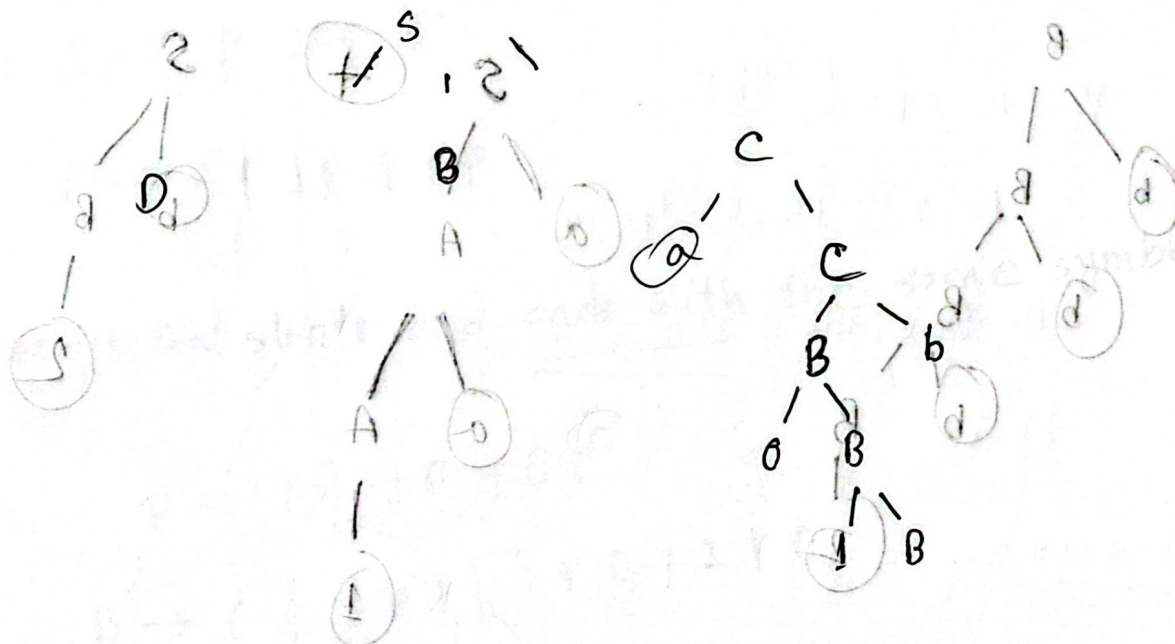
LEAS
11
(d)

switch (5): (INT) No figure <= HIGH

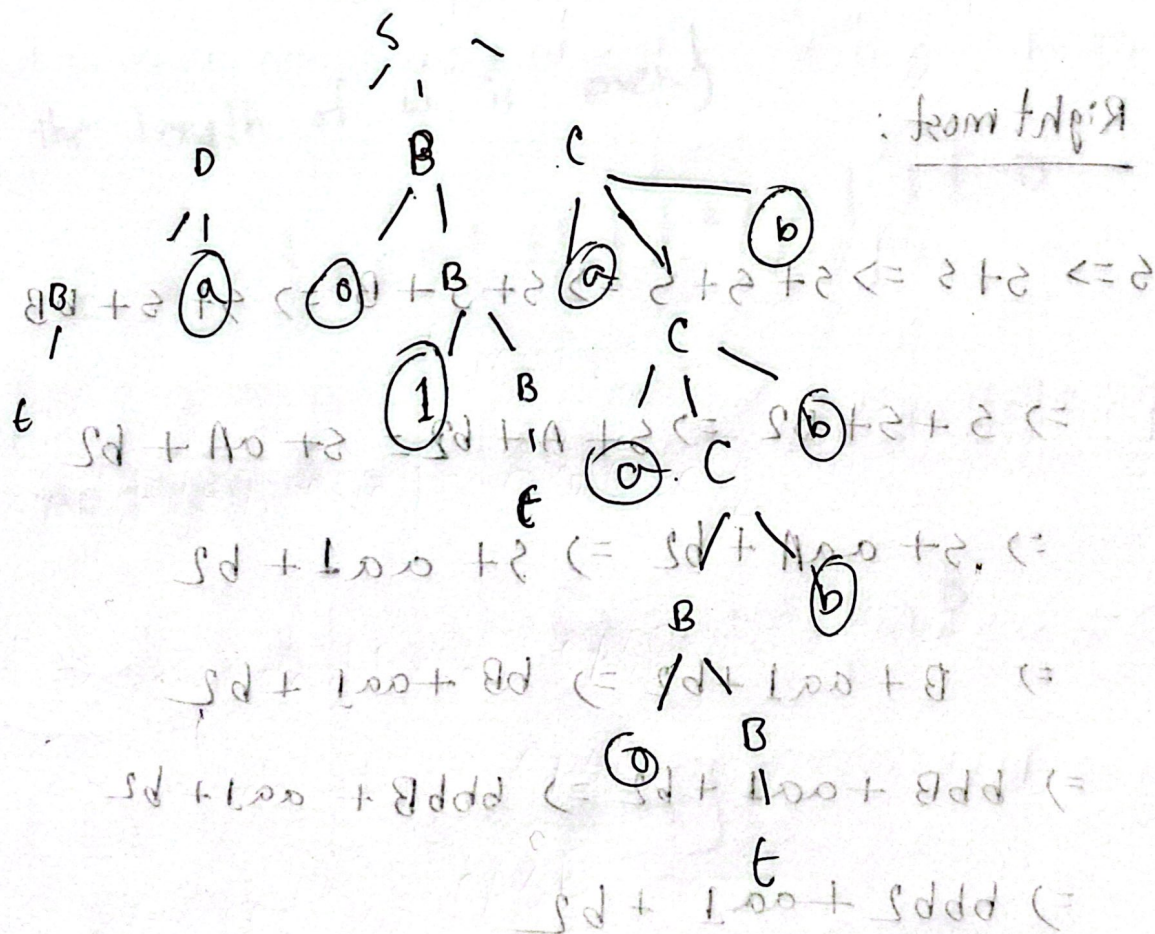
switch (5): (INT) No figure <=

switch (5): (INT) No figure

a01a0bbb



Right most:



242:
2

$$(a) L = \{ a^n b^{2m} c^n \mid n, m \geq 1 \}$$

$$A \rightarrow aAc \mid aBc$$

$$B \rightarrow bB \mid b$$

(b)

$$P \rightarrow a \mid b \mid aPa \mid bPb$$

$$(d) L = \{ p^n q^m r^0 s^x \mid n = 2m, x = 20, n, x \geq 0 \}$$

$$p^{2m} q^m r^0 s^{20}$$

$$S \rightarrow AB$$

$$\begin{array}{l} A \rightarrow \epsilon \mid pPA \\ A \rightarrow \epsilon \mid pPA \\ B \rightarrow \epsilon \mid rBss \end{array}$$

$$S \rightarrow AB$$

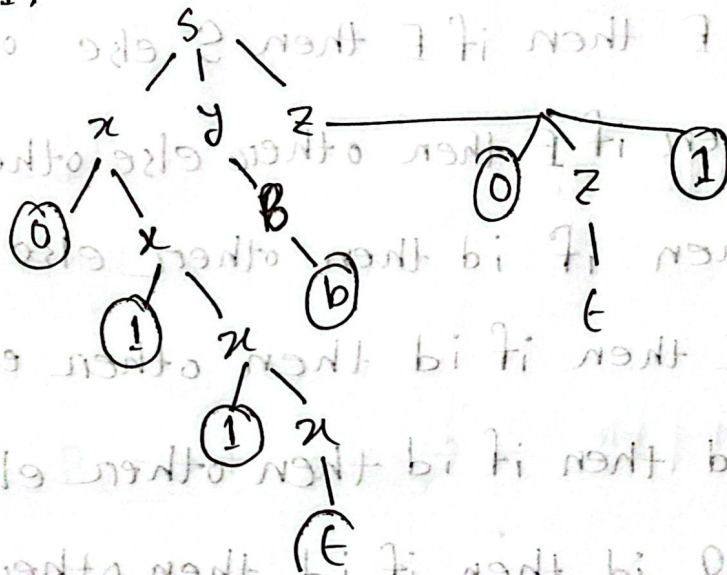
$$E \rightarrow \epsilon \mid aE \mid pEP$$

$$O \rightarrow a \mid p \mid aOa \mid pOp$$

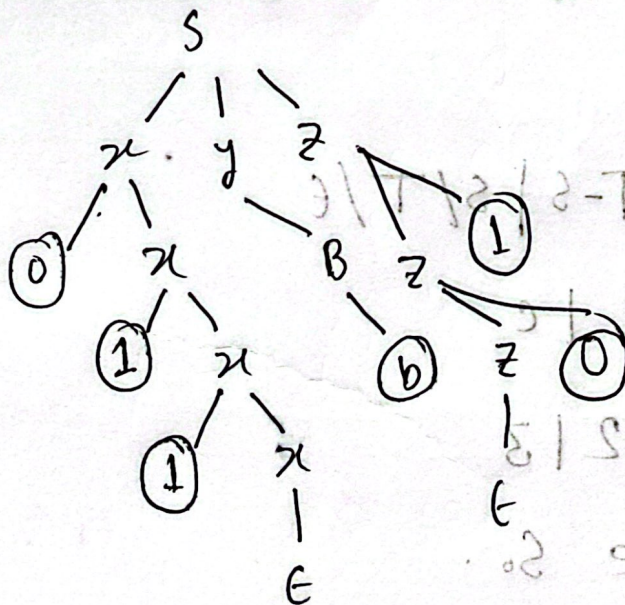
(a)

011b01

Tree 1:



Tree 2:



(Ambiguous.)

(b)

$S \xRightarrow{R_m} \text{if } E \text{ then } S \text{ else } S$

$\Rightarrow \text{if } E \text{ then } S \text{ else other}$

$\Rightarrow \text{if } E \text{ then if } E \text{ then } S \text{ else other}$

$\Rightarrow \text{if } E \text{ then if } E \text{ then other else other}$

$\Rightarrow \text{if } E \text{ then if id then other else other}$

$\Rightarrow \text{if } E \ \&\& \ E \text{ then if id then other else other}$

$\Rightarrow \text{if } E \ \&\& \ \text{id then if id then other else other}$

$\Rightarrow \text{if } E \parallel E \ \&\& \ \text{id then if id then other else other}$

$\Rightarrow \text{if id} \parallel \text{id} \ \&\& \ \text{id then if id then other else other}$

3

(a) $S \rightarrow S + S \mid T - S \mid S / T / \epsilon$

$T \rightarrow T * TC \mid C$

$C \rightarrow 0 \mid 1 \mid 2 \mid 3$

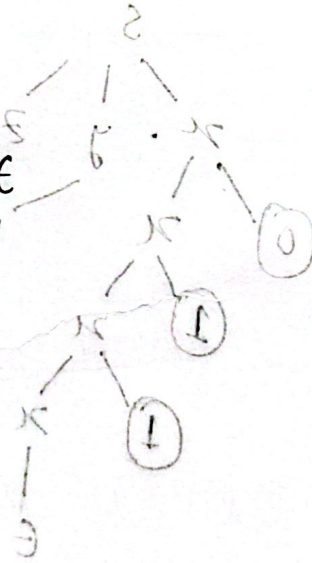
Let start variable S_0 .

$S_0 \rightarrow S$

$S \rightarrow S + S \mid T - S \mid S / T \mid \epsilon$

$T \rightarrow T * TC \mid C$

$C \rightarrow 0 \mid 1 \mid 2 \mid 3$



(.2voipidmH)

remove ϵ :

$$S_0 \rightarrow S+S \mid T-S \mid S/T \mid S+ \mid +S \mid + \mid T-A \mid 0/T A$$

$$S \rightarrow S+S \mid T-S \mid S/T \mid S+ \mid +S \mid + \mid T- \mid /T$$

$$T \rightarrow T * T C \mid C$$

$$C \rightarrow 0 \mid 1 \mid 2 \mid 3$$

$$A \rightarrow TC$$

$$B_1 \rightarrow * , B_2 \rightarrow + , B_3 \rightarrow - , B_4 \rightarrow /$$

$$P \rightarrow T$$

$$Q \rightarrow P B_1$$

$$R \rightarrow Q A$$

$$D_0 \rightarrow 0$$

$$D_1 \rightarrow 1$$

$$D_2 \rightarrow 2$$

$$D_3 \rightarrow 3$$

$$S_0 \rightarrow S$$

$$S \rightarrow S B_2 S \mid P B_3 S \mid S B_4 P \mid S B_2 \mid B_2 S \mid B_2 \mid P B_3 \mid B_4 P$$

$$P \rightarrow R$$

$$C \rightarrow D_0 \mid D_1 \mid D_2 \mid D_3$$

$$(b) \quad S \rightarrow SS \mid CAB \mid BA$$

$$A \rightarrow aAb \mid \epsilon$$

$$B \rightarrow oA1 \mid oB1$$

$$C \rightarrow cA \mid Bb$$

Let, starting Variable S_0 .

$$S_0 \rightarrow S$$

$$S \rightarrow SS \mid CAB \mid BA$$

Remove ϵ from S .

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow S \mid CAB \mid BA$$

$$\Rightarrow S_0 \rightarrow S \mid CAB \mid BA$$

$$S \rightarrow S \mid CAB \mid BA$$

$A -$

Remove ϵ from A .

$$A \rightarrow Ab \mid aAb$$

$$B \rightarrow o1 \mid oB1 \mid oA1$$

$$C \rightarrow cA \mid Bb \mid c$$

$$S_0 \rightarrow S \mid A_0 B \mid BA$$

$$A_0 \rightarrow cA$$

$$S \rightarrow S \mid A_0 B \mid BA$$

$$A \rightarrow D \mid ED_3$$

$$D \rightarrow ab$$

$$D_1 \rightarrow a$$

$$D_2 \rightarrow A$$

$$D_3 \rightarrow b$$

$$E \rightarrow D_1 D_2$$

$$F \rightarrow ED_3$$

$$B \rightarrow B_1 \mid aB_3 \mid HB_3$$

$$B_1 \rightarrow 01$$

$$B_2 \rightarrow 0$$

$$B_3 \rightarrow 1$$

$$G_1 \rightarrow 0B$$

$$H \rightarrow 0A$$

$$C \rightarrow cA \mid BD_3 \mid c_1$$

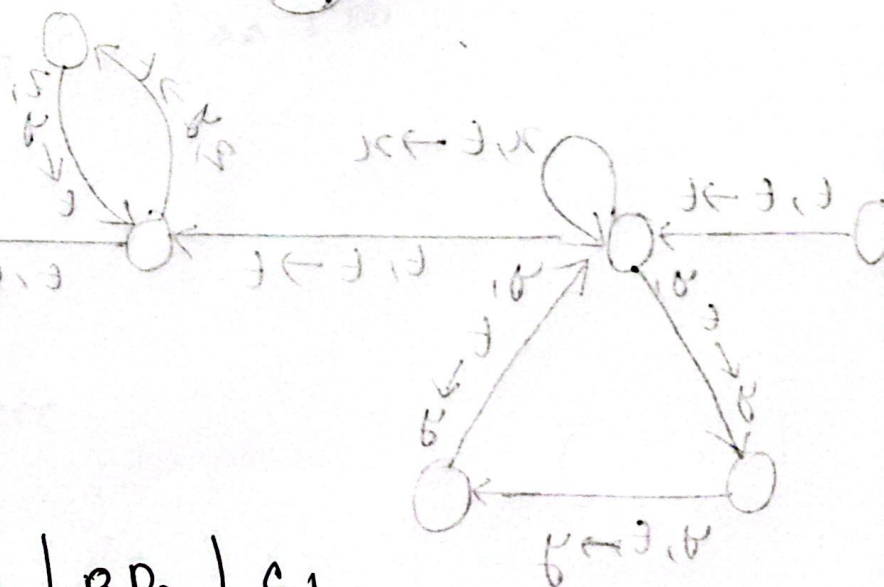
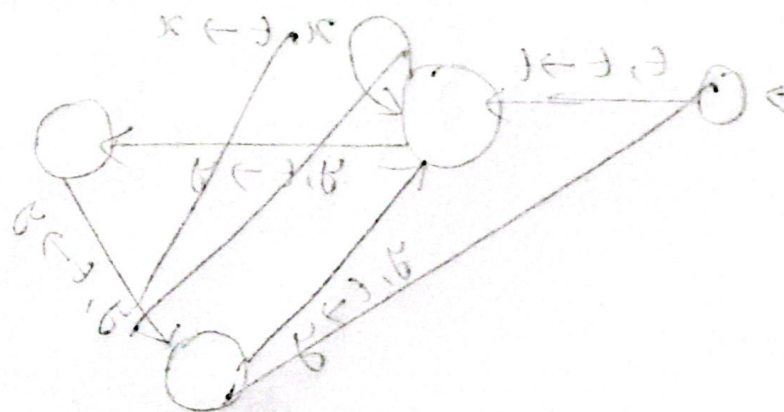
$$c_1 \rightarrow c$$

$$(x) \rightarrow (y) \quad \text{broad}$$

$$(y) \rightarrow (x) \quad \text{broad}$$

$$(y) \rightarrow (x) \quad \text{broad}$$

$$(x) \rightarrow (y) \quad \text{broad}$$



Summer -24:

4) (a) $L = \{ x^n y^{3m} z^{2m} v^n \mid v, m \geq 1 \}$

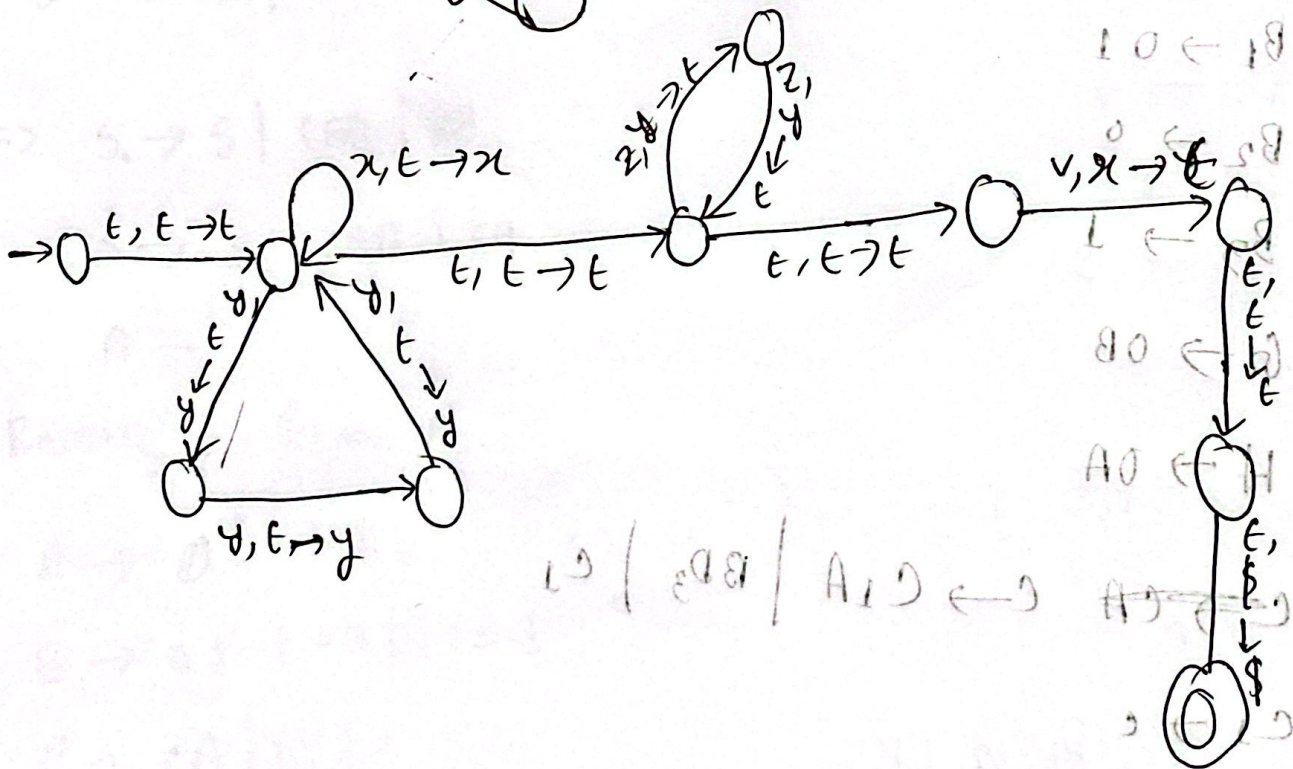
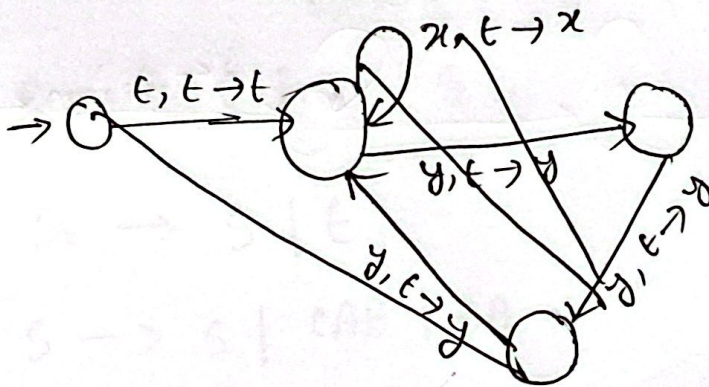
algorithm:

read $(x) \rightarrow \text{push}(x)$

read $(y) \rightarrow \text{push}(y)$

read $(z) \rightarrow \text{pop}(y)$

read $(v) \rightarrow \text{pop}(x)$



Fall-24:

4.

(b) $3^i \quad 2^j \quad 2^k$
 $p \quad \# \quad q \quad r$

$i = j$

$3^i \quad 2^i \quad 2^k$
 $p \quad \# \quad q \quad r$

read (3p) $\rightarrow$ push (p)

read (#) $\rightarrow$ no change

read (2q) $\rightarrow$ pop (1p)

read (r) $\rightarrow$ do nothing

$\downarrow$
 check $\rightarrow$ if r is even or not.

$i = k$ $3^i \quad 2^j \quad 2^i$
 $p \quad \# \quad q \quad r$

read (3p) $\rightarrow$ push (p)

read # $\rightarrow$ no change

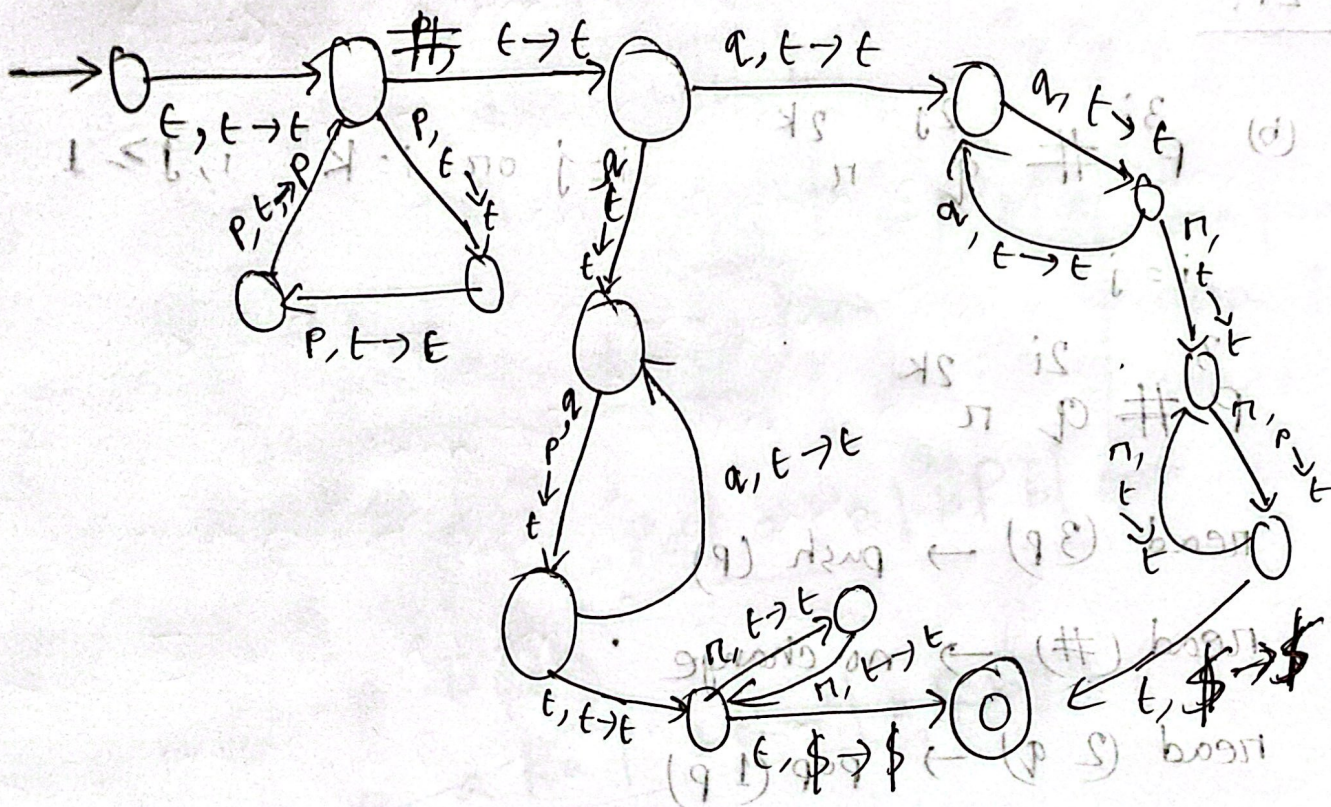
read even(q) $\rightarrow$ no change

read (2r) $\rightarrow$ pop 1p

2809/10/25
(bom)

507

PS-1101



printing ab $\leftarrow$ (15) begin

check - if it is even or not

$i = k$
 $b \neq \#$
 $s \neq \#$
 $i \neq \#$

used (36) $\leftarrow$ begin (6)

used $\#$ - no change

used even (4) - no change

used (51) $\leftarrow$ for 16

243:

$$4) L = \{ x^a y^b z^c w^d \mid (c=3b \text{ and } a=2d) \text{ and } a \geq 1, b \geq 0 \}$$

$\begin{matrix} 2d & 2b & 3b & d \\ x & y & z & w \end{matrix}$

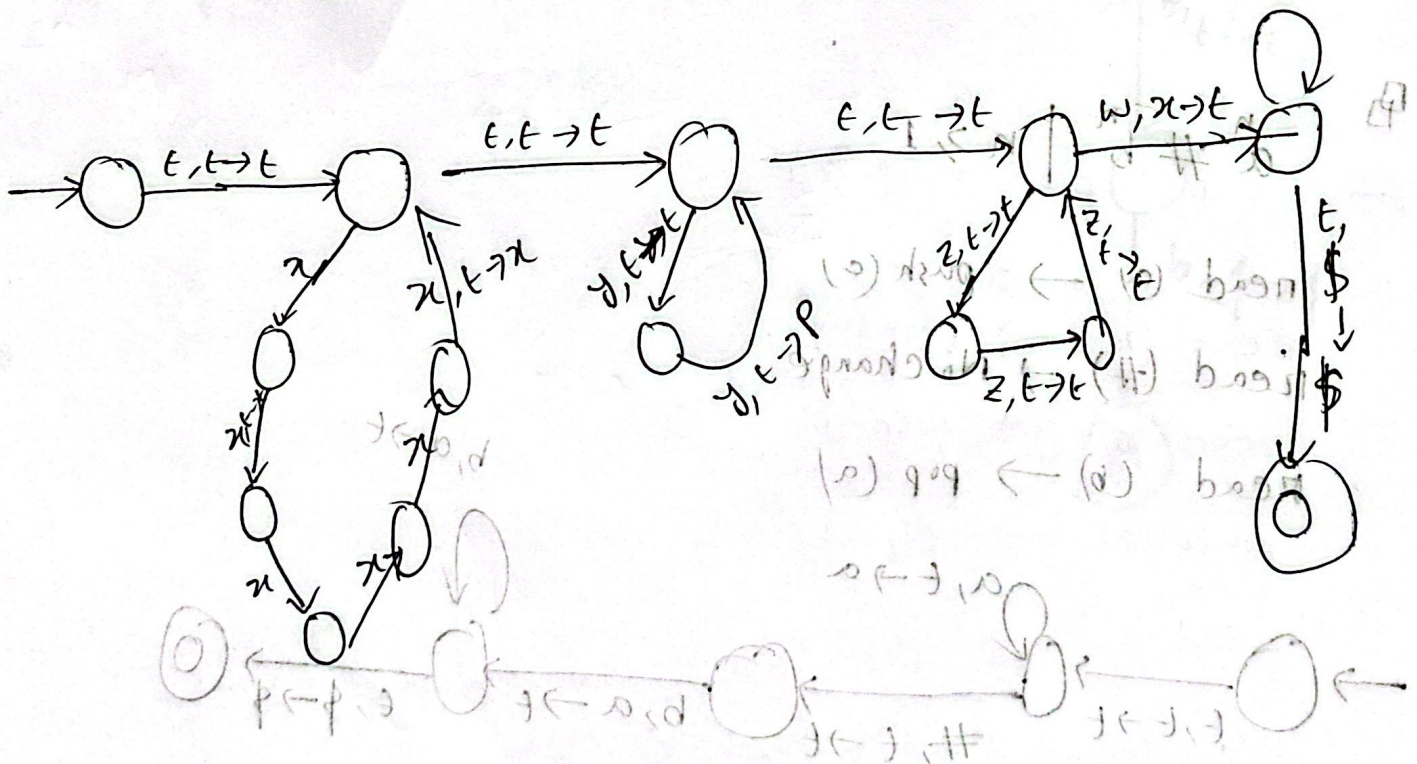
$$[c=3b \text{ and } a=2d]$$

(i) read (1x) $\rightarrow$ push 1(x)

(ii) read (2y) $\rightarrow$ push 1(y)

(iii) read (3z) $\rightarrow$ pop 1(y)

(iv) read (1w) $\rightarrow$ pop 1(x)



Fall 2024:

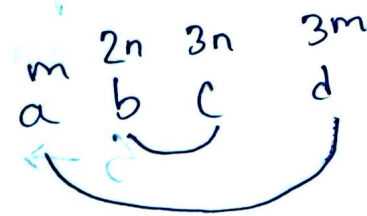
(2) (a) $L = \{ a^n b^m \mid n \leq m \leq 2n \text{ and } n, m \geq 1 \}$

$S \rightarrow ab \mid abb \mid asb \mid asbb \mid \dots$

(b) $L = \{ a^m b^{2n} c^{3n} d^{3m} \mid \text{where } n \geq 2, m \geq 1 \}$

$S \rightarrow aSddd \mid aAddd$

$A \rightarrow bbbccccc \mid bbAccc$



$n = 2, 3, 4, \dots$
 $m = 1, 2, 3, \dots$

best $\rightarrow$
 $A \rightarrow aBddd \mid aAddd$
 $B \rightarrow bbbbccccc$

$$(c) L = \{ a^i b^j c^k \mid \text{where } i=k \text{ or } j=k \text{ and } i, j, k \geq 0 \}$$

$$a^i b^j c^i \quad [i=k] \quad \text{or} \quad a^i b^j c^j \quad [j=k]$$

$$S \rightarrow aSc \mid aBc$$

$$B \rightarrow b \mid bB$$

$$R \rightarrow CA$$

$$A \rightarrow bAc \mid bc$$

$$C \rightarrow aC \mid a$$

$$P \rightarrow S \mid R$$

$$S \rightarrow aSc \mid aBc$$

$$B \rightarrow b \mid bB$$

$$R \rightarrow CA$$

$$A \rightarrow bAc \mid bc$$

$$C \rightarrow aC \mid a$$

$$(d) S \rightarrow aEa \mid bOb \mid b$$

$$E \rightarrow \epsilon \mid aEa \mid bEb$$

$$O \rightarrow a \mid b \mid aOa \mid bOb$$